Journal of Ethnopharmacology 198 (2017) 205–213

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jep

# Ameliorating effects of herbal formula hemomine on experimental subacute hemorrhagic anemia in rats

Sae Kwang Ku[a,1], Hyemee Kim[b,1], Joo Wan Kim[c], Ki Sung Kang[d], Hae-Jeung Lee[e,*]

[a] Department of Anatomy and histology, College of Korean Medicine, Daegu Haany University, Gyeongsan, South Korea
[b] Department of Nutrition and Food Science, Texas A & M University, College Station TX, United States of America
[c] College of Veterinary Medicine, Kyungpook National University, Daegu, South Korea
[d] College of Korean Medicine, Gachon University, Seongnam-si, Gyeonggi-do, South Korea
[e] Department of Food and Nutrition, Gachon University, Seongnam-si, Gyeonggi-do, South Korea

ARTICLE INFO

*Keywords:*
Anemia
Hematopoietic herbal extracts
Digestive discomfort
Colonic mucosa

ABSTRACT

*Ethnopharmacological relevance:* Hemomine (HM) is an herbal mixture consisting of 5 varieties of the hematopoietic herbal extracts (*Angelica gigas* Nakai, *Cnidium officinale* Makino, *Paeonia lactiflora* Pall., *Rehmannia glutinosa* Liboschitz ex Stueudel, *Glycyrrhiza uralensis* Fischer).

*Aim of the study:* Anemia has been treated with iron supplements, whereas it could cause adverse side effects such as digestive discomfort. In the present study, HM was applied to SHA rats to test for several activities so as to verify its therapeutic potentials on anemia and digestive discomfort.

*Materials and methods:* Sprague-Dawley rats were assigned to seven groups: (Two controls, two references (ferric hydroxide polymatose (FM) and ferritin extract glycerin hydrate (FA)), three different concentrations of HM, n=8 per groups), and induced subacute hemorrhagic anemia (SHA) through blood exsanguinations once a day for 7 days.

*Results:* The SHA animal model showed changes in the markers related to classic iron-deficient and regenerative anemia in this experiment. However, the SHA related anemic signs were dose-dependently inhibited by the administration of HM 2, 1, and 0.5 ml/kg for 7 days, and more favorably than the equal dosages of FM and FA. In addition, FM and FA showed the typical constipation signs, including reduction of in thickness of the colonic mucosa, in contrast, HM 2, 1, and 0.5 ml/kg groups had no effects on the gastrointestinal motilities and the colonic mucous components when compared to the controls. The results suggested that the HM significantly showed to have therapeutic effects in the experimental SHA in rats, and is more potent than the commercial iron supplement through the proliferation of hematopoietic stem cells with reduced digestive discomfort.

*Conclusions:* Therefore, Hemomine may prove to be a promising hematopoietic and therapeutic agent for anemia.

## 1. Introduction

Global prevalence of anemia was 32.9% and is most prevalent among children $<5$ years and women (Kassebaum et al., 2014). Iron deficiency anemia is the most common disorder of anemia (low red blood cells or hemoglobin levels) caused by insufficient dietary intake, altered iron absorption, or iron depletion from bleeding (Cook, 2005). Iron deficiency anemia has been treated by iron supplements; however, the treatments have been reported to cause adverse side effects, for example, digestive discomfort (Sahebzamani et al., 2013; Souza et al., 2009). Although treatments with ferric hydroxide polymaltose complex or ferritin extract glycerin hydrate has a lower incidence of adverse events than iron (II) sulfate, they also induced various symptoms of digestive discomfort, such as constipation (Geisser, 2007; Szarfarc et al., 2001). Therefore, new strategies are highlighted as a modern approach to seek out potent anti-anemic agents with reduced toxic side effects, rather than the conventional iron supplements, especially those concerned with natural herbs or their mixtures (Akase et al., 2003; Feltrin et al., 2009; Nakamoto et al., 2008).

Hemomine (HM) is an herbal mixture (at least 60% solids, LJG305)

* Correspondence to: Department of Food and Nutrition, Gachon University, 1342 Seongnam-daero, Sujeong-gu, Seongnam-si, Gyeonggi-do 13120, South Korea.
*E-mail addresses:* gucci200@hanmail.net (S.K. Ku), kimhyemee@tamu.edu (H. Kim), warii@hanmail.net (J.W. Kim), kkang@gachon.ac.kr (K.S. Kang), skysea@gachon.ac.kr, skysea1010@gmail.com (H.-J. Lee).
[1] These authors contributed equally to this work.

http://dx.doi.org/10.1016/j.jep.2017.01.010
Received 23 September 2016; Received in revised form 23 December 2016; Accepted 9 January 2017
Available online 09 January 2017
0378-8741/ 

consisting of 5 varieties of the hematopoietic herbal extracts including: roots of *Angelica gigas* Nakai, rhizomes of *Cnidium officinale* Makino, roots of *Paeonia lactiflora* Pall., roots of *Rehmannia glutinosa* Liboschitz ex Stueudel, and roots of *Glycyrrhiza uralensis* Fischer. Roots of *Angelica gigas* Nakai is the major material in HM and rich in bioactive compounds, including nodakenin, decursin, and decursinol angelate (Lee et al., 2003). Among the compounds, nodakenin was reported to have anti-inflammatory (Rim et al., 2012), anti-allergic (Kim et al., 2011), anti-obesity activities (Cha et al., 2011), and memory-enhancing properties (Kim et al., 2007). The potential use of individual hematopoietic herbal extracts of HM have been well documented: *Angelica gigas* Nakai (Kim et al., 2013), *Paeonia lactiflora* Pall. (Son et al., 2003), and *Rehmannia glutinosa* Liboschitz ex Stueudel (Xu, 1992). Some mixed herbal formulas containing *Cnidium officinale* Makino (Jo et al., 2005) or *Glycyrrhiza uralensis* Fischer (Zee-Cheng, 1992) were reported to have the anti-anemic effects. In addition, the following vitamins: Vitamin B1, B2, B6, B12, and folic acid are essential for the process of erythropoiesis and are included in HM (Koury and Ponka, 2004).

Subacute hemorrhagic anemia (SHA) is a specific type of anemia that denotes a precipitous drop in red blood cells levels and iron due to acute hemorrhage. Also, it is easily induced in rats by continuously exsanguinating blood in considerable amounts and frequencies, about 1–2 ml of blood once a day for a period of 5–7 days (Pulina et al., 2010; Younes et al., 2007); therefore it has been used for screening the efficacy of hematopoietic agents or iron supplements (Piao et al., 2003; Pulina et al., 2010; You et al., 1996).

The purpose of the study is to investigate the possible alleviating effects of HM, a hematopoietic herbal mixture, on the experimental SHA in rats. In addition, the anti-anemic effects of HM were compared to the commercial iron supplements.

## 2. Materials and methods

### 2.1. Preparations of test materials

FM (7.14% ferric hydroxide polymaltose complex: Feromax™, Hanmi Pharm. Co. Ltd., Seoul, Korea), FA (20% ferritin extract glycerin hydrate: Fematin-Asyrup™, Cho-A Pharm. Co. Ltd., Haman, Korea), and HM (Hemomine, 60% LJG305) were supplied by Aribio Inc. Ltd. (Jechon, Korea). All test materials were stored at 4 °C in a refrigerator to protect it from the light and humidity until the extracts were utilized. HM solutions were diluted in distilled water as 1:9, 1:8, and 1:6 as finalized doses were 0.5, 1, and 2 ml/kg. FM solution was also diluted in distilled water as 1:9, and FA solution as 1:8 (finalized dosages were 0.5 ml/kg of FM and 1 ml/kg of FA). The FM and FA groups' doses were selected based on previous human clinical trials (Coyne et al., 2007; Geisser, 2007). The medium and low dosages of HM groups were selected as 1 and 0.5 ml/kg to match the doses for the FM and FA groups to compare their anti-anemic effects. 2 ml/kg of HM was selected as a high dose using the common ratio 2 to see if the HM need a higher dose than the current medications.

### 2.2. Hemomine chemistry

Hemomine has a proprietary blend of roots of *Angelica gigas* Nakai (3.9%), rhizomes of *Cnidium officinale* Makino, roots of *Paeonia lactiflora* Pall., roots of *Rehmannia glutinosa* Liboschitz ex Stueudel, and roots of *Glycyrrhiza uralensis* Fischer. Characterization of bioactive compounds in Hemomine was performed on an Agilent 1260 Infinity (Agilent, Palo Alto, CA), which was used to 10 µl aliquots of the processed samples onto a Hector C18 column (250 mm x 2.0 mm, 5 µm) equilibrated at 30 °C. The gradient mobile phase, with solvent A (100% acetonitrile, v/v) and solvent B (100% water, v/v) were set at a flow rate of 1 ml/min. Nodakenin as the standard was obtained from Chemface Co. (Wuhan, China). HPLC grade acetonitrile was obtained

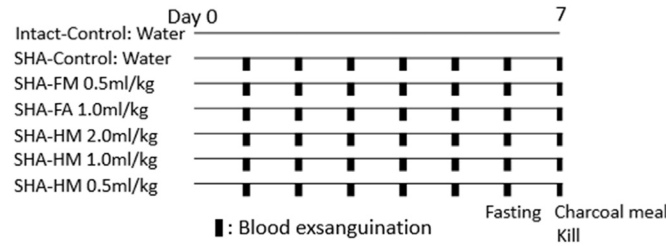


**Fig. 1.** Experimental design of the study. A total of seven groups of rats was studied (n=8 per group). In the intact control rats, blood exsanguinations did not performed, whereas in the subacute hemorrhagic anemia (SHA) groups, anemia were induced by exsanguinations of blood (1 ml) from orbital plexus, once a day for 7 days. The animals were fasted overnight 18 h before they were killed. FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), HM: Hemomine (60% LJG305).

from Burdick & Jackson (Muskegon, MI).

### 2.3. Study design and induction of SHA

This study was approved by Institutional Animal Care and Use Committee at Daegu Haany University (DHU2014-002). A total sample size of 56 male Sprague-Dawley rats (6 week-old, 150–170 g) were purchased from OrientBio (Seongnam, Korea) and maintained in polycarbonate cages in a room controlled at 20–25 °C, 50–55% humidity, and a 12 h light/dark cycle. Rats had free access to standard rodent chow (Samyang, Korea) and water. After the rats acclimated to their environment for a period of 8 days, the rats were allocated to seven groups (n=8 per group, Fig. 1). SHA was induced according to previous studies (Pulina et al., 2010; Younes et al., 2007). Rats were anesthetized with 2–3% isoflurane (Hana Pharm. Co., Hwasung, Korea) in a mixture of 70% $N_2O$ and 28.5% $O_2$. Blood exsanguinations (1 ml/head, match to about 0.5% of body weights) were continuously executed from the orbital venous plexus, once per day for 7 consecutive days under isoflurane anesthesia by a skilled veterinarian at one hour before the test material was administered. Orbital plexus puncture is a survival rodent bleeding technique used frequently in rats for rapid collection of large blood volumes (Sharma et al., 2014). In addition, the SHA model induced by blood draw from orbital plexus is the most rapid effective in producing the clinical features of anemia including high reductions in hemoglobin, serum iron, and red blood cells in experimental animal (Ihedioha and Daniel-Igwe, 2014; Pulina et al., 2010). Even though it is recommended that a cumulative collection of 10% of total blood volume (1.3 ml/230 g rat) not be exceeded for a week period (Diehl et al., 2001; Lee and Blaufox, 1985), the blood draw procedure is necessary to induce the SHA model. In addition, no adverse effects were reported in the previous studies (Ihedioha and Daniel-Igwe, 2014; Pulina et al., 2010). Therefore, the animal ethics committee approved the location (orbital venous plexus), frequency (once a day for 7 days), and volume (1 ml/head, 0.5% of body weight) of the blood draw procedure. In the intact and SHA control rats, distilled water (5 ml/kg) was orally administered. Humane endpoints were in place during the study following Canadian Council on Animal Care and ICLAS, Harmonization of guidelines on the use of animals in science (Demers et al., 2006). The criteria for euthanasia were over 20% of body weight loss, hemorrhagic shock, hypersensitiveness, tachypnea and labored respiration, and moribund condition (Demers et al., 2006). No adverse events were occurred and no mortality was observe prior to the endpoint during the study. In the intact vehicle control rats, no exsanguinations were conducted, but the system inhalation anesthesia was performed. Each group was orally administrated in a volume of 5 ml/kg, as finalized dosages were 0.5 ml/kg of FM, 1 ml/kg of FA, and 2, 1, and 0.5 ml/kg of HM, once a day for 7 days at one hour apart from blood exsanguinations. Animals were monitored and body weight was recorded daily. The FM and FA medications were reported to have side effects of constipation. Fasting prior to termination followed by feeding charcoal meal to the rats is the standard way to test for constipation as a result of these medications. Therefore the animals were fasted overnight 18 h before they were killed. At the time of necropsy, 0.5 ml of blood were collected from the orbital plexus while inhaling anesthesia. The rats were euthanized by $CO_2$, and the organs including the spleen, liver, and left femur were weighed and documented.

### 2.4. Hematology and smear cytology

All hematological measurements were conducted in Veterinary Teaching Hospital, College of Veterinary Medicine, Kyungpook National University (Daegu, Korea) using automated hematology cell counter (MS9-5V, Melet Schloesing Lab., Paris, France). The following markers of hematology were measured: leukocytes (WBC), erythrocytes (RBC), hemoglobin (Hb), hematocrit (HCT), micro-erythrocyte (small erythrocyte to normal range, µRBC) and macro-erythrocyte (larger erythrocyte to normal range, MRBC), and platelets (PLT). At the time of necropsy, the collected blood was directly smeared onto the slides. The slides were fixed by submerging the slides in absolute methanol for 30 min and stained by 1:6 diluted Giemsa (Sigma-Aldrich, St. Louise, MO, USA) for 10 min. The slides were randomly coded and examined under a ×400 magnification by two different experts. RBC diameters (µm/cell) and polychromatic erythrocytes (PCE/1000 cells) were measured using a computer-assisted imaging analysis program, iSolution FL ver 9.1 (IMT i-solution Inc., Vancouver, Canada) to conclude which type of anemia was present: a microcytic anemia, macrocytic anemia, regenerative anemia, or degenerative anemia (Janus and Moerschel, 2010; Price et al., 2011).

### 2.5. Histopathology

The collected spleen, left lateral lobes of the liver, and left femur were fixed in a 10% neutral buffered formalin. After fixation, the femur samples were decalcified (24.4% formic acid and 0.5N sodium hydroxide, replaced every day) for five days, however, the liver and spleen samples were not fixed. Next, the femur samples were embedded in paraffin then were sectioned (3–4 µm) and stained with Hematoxylin & Eosin (H & E). The total number of splenic red pulp cells ($\times 10^3$ cells/$mm^2$), area of hematopoietic spots in the liver parenchyma (%/$mm^2$), and the total number of femur bone marrow cells ($\times 10^3$ cells/$mm^2$) were measured using a computer-assisted image analysis program, iSolution FL ver 9.1 (IMT i-solution Inc., Vancouver, Canada).

In addition, assessment of the histological observations of the colon mucosa and fecal pellets remnant in the lumen of the colon were determined to observe digestive disorders, particularly constipation, which is the major discomfort side effect caused by iron supplements intake (Sahebzamani et al., 2013; Souza et al., 2009). Segments of the rat's distal colon containing one fecal pellet were isolated by ligatures, removed, and immediately fixed with 10% formaldehyde for the intestinal charcoal transit ratio measurement. The fixed tissue segments were embedded in paraffin and serially cut measuring 3 µm thick. The sections were stained with alcian blue (pH 2.5). The histological profiles were measured as colonic mucosa thicknesses (µm), mucosal goblet cell (alcian blue positive cells /$mm^2$ of colonic mucosa), and the thickness of mucous layer on the fecal surface (µm) using a computer-assisted image analysis program, iSolution FL ver 9.1 (IMT i-solution Inc., Vancouver, Canada) (Wu et al., 2011). The histopathology markers were measured blindly.

### 2.6. Immunohistochemistry

Changes of CD34 and CD45-immunoreactive cells, which are hematopoietic stem cells, were observed by immunohistochemistrical methods using CD34 and CD45 antibodies (BD Biosicences, San Jose, CA, USA) with avidin-biotin-peroxidase (ABC) and the peroxidase substrate kit (Vector Labs, Burlingame, CA, USA). Endogenous peroxidase activity was blocked *via* incubation in methanol and 0.3% $H_2O_2$ for 30 min. The sections of the spleen and liver samples were heated (95–100 °C) for epitope retrieval in 10 mM citrate buffers (pH 6.0) (Shi et al., 1993), and the sections of femur samples were treated with trypsin (Sigma-Aldrich, St. Louise, MO, USA) and 2N HCl (Tang et al., 2007). Specific binding was blocked by using normal horse serum blocking solution for one hour in an humidity chamber. The primary antibodies were treated overnight at 4 °C, and then the sections were incubated with biotinylated universal secondary antibody and ABC reagents for one hour at room temperature in the humidity chamber. Finally, the slides were reacted with the peroxidase substrate kit for 3 min at room temperature. All the sections were rinsed in 0.01 M Phosphate buffered saline three times between each step. If the cells were occupied by over 20% of the immunoreactivities, the results were positive. The number of CD34 and CD45-immunoreactive cells in the spleen, liver, and the femur bone marrow were counted using a computer-assisted image analysis program, iSolution FL ver 9.1 (IMT i-solution Inc., Vancouver, Canada). The immunohistochemistry markers were measured blindly.

### 2.7. Measurement of intestinal charcoal transit ratio

Gastrointestinal propulsion of the charcoal meal was assessed (Sagar et al., 2005). The animals tested were starved for 18 h before the experiment but consumed water *ad libitum*. Ten minutes later of last 7th test material administration, the animals were fed 1 ml of the charcoal meal (3% suspension of activated charcoal in 0.5% aqueous methylcellulose (Sigma-Aldrich, St. Louise, MO, USA)). Thirty minutes after the administration of the charcoal meals, the animals were killed, and the total length of the intestine (pyloric sphincter to cecum), as well as the distance along which the charcoal moved, were measured. The intestinal charcoal transit ratio was calculated as the difference between the lengths of small intestine and charcoal meal transferred.

### 2.8. Statistical analyses

All data were expressed as Mean ± standard deviation (SD) (n=8 per group). Multiple comparison tests for the different dose groups were conducted. If the Levene test indicated an absence of significant deviations from the variance homogeneity (Levene, 1981), the obtained data was analyzed by the one-way ANOVA test and the least significant differences multi-comparison (LSD) test to determine which pairs of group comparisons were significantly different. If the Levene test indicated the presence of significant deviations from the variance homogeneity, a non-parametric comparison test (Kruskal-Wallis and Mann-Whitney test) was conducted. Statistical analyses were conducted using SPSS (Release 14.0 K, IBM SPSS Inc., Armonk, NY, USA).

## 3. Results

### 3.1. Bioactive compounds in Hemomine extract

Using chromatographic analysis of Hemomine extract, nodakenin was identified (0.032 mg/ml in Hemomine extract) with peak absorption at 330 nm (Fig. 2). The rats in the HM groups were fed 16–62 µg/kg b.w. of nodakenin per day.

### 3.2. Changes in body and organ weights

The SHA control, SHA-FM and SHA-FA showed significant decreases of the body weight from 4 days after exsanguinations and the final body weight gain after 7 days compared to the intact control. There were no significant changes on the body weight and the weight gain in the FM and FA groups in comparison to the SHA control during

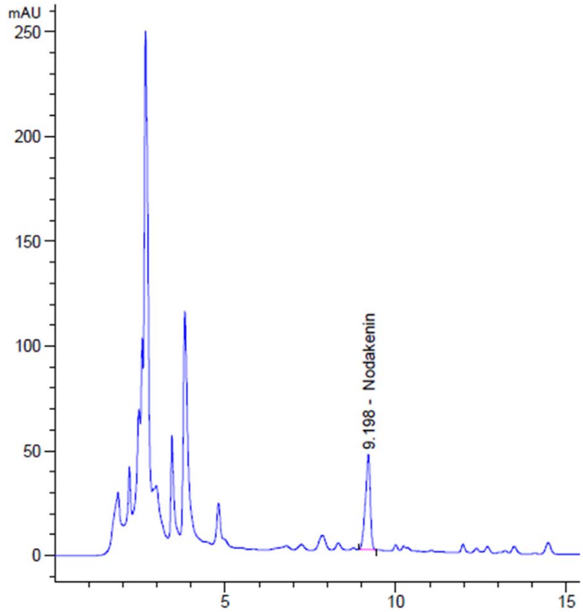


**Fig. 2.** Representative chromatographic profile at 330 nm of bioactive compounds in Hemomine.

the study. However, HM 2 and 1 ml/kg treated rats showed significant increase in body weight from 4 days after administration, and all three different dosages of HM 2, 1, and 0.5 ml/kg treated rats showed significant increase in body weight gain by 126.44%, 85.63%, and 57.47%, respectively, after 7 days of administration periods, compared to the SHA control rats ($p < 0.05$). There was no significant difference in the body weight gains between the intact control and the HM 2 ml/kg group (Table 1). The observed maximum percentage weight loss was −2.75% per day during the study (One rat in the SHA-FA group between day 5 and 6). The rat showed the lowest body weight gain (13.73%; average 18.19% in the FA group) during the study.

The relative spleen weights in the SHA control, SHA-FM, SHA-FA, and SHA-HM 0.5 ml/kg, were significantly increased by 38.38%, 39.96%, 50.45%, and 19.36%, respectively when compared to the intact control, and it was reduced in the HM 2, 1, and 0.5 ml/kg treated rats. The relative liver weight in the SHA control was significantly decreased by −7.41% in comparison to the intact control, and observed a significant increase in the HM 2 ml/kg treated rats when compared to the SHA control. There were no significant changes in the relative femur weights (Table 1).

### 3.3. Changes in hematology and blood smear cytology

Significant decreases in RBC, Hb, and HCT, and increases in μRBC, MRBC ratios, and PLT in the SHA control rats in comparison to the intact control ($p < 0.05$) were observed. However, the noted abnormal changes in the hematological inspections were recovered by treatment of all the test substances, including two commercial iron supplements, with the exception of the levels of the MRBC and PLT. Particularly, HM 2 and 1 ml/kg treated rats showed more favorable and significant normalizations in hematological values in comparison to the FM and FA treated rats ($p < 0.05$). There were no notable changes in the level of WBC (Table 2).

In addition, there was a significant reduction in the RBC diameters by −30.78%, and an induction of the PCEs among 1000 RBCs by 422.65% in the SHA control rats compared to the intact control. However, these abnormal changes in the cytological smear inspections were significantly normalized by the treatment of all the test substances, including FM and FA. HM 2 and 1 ml/kg treated rats showed more favorable and significant normalization of RBC sizes and proportions of PCEs in comparison to the FM and FA treated rats ($p < 0.05$) (Table 2, Fig. 3A).

### 3.4. Changes in spleen, liver, and femur bone marrow histopathology

There were noticeable inductions of splenic red pulps, hematopoietic spots suggesting ectopic hematopoiesis in the liver, and hematopoietic cell proliferations in the marrow of the femur bone in the SHA control rats compared to the intact control ($p < 0.05$). In addition, there were significant decreases in the CD34+ and CD45+ immune cells in the spleen, liver, and femur bone marrow of the SHA control rats when compared to the intact control. However, all the test treatments significantly reduced the number of red pulp cells in spleen, the area of hematopoietic spots in liver, and the number of femur bone marrow cells. However, an increase in the CD34+ and CD45+ cells in spleen, liver, and femur bone marrow, compared to the SHA control rats, were observed. Especially, HM 2 and 1 ml/kg treatment showed more favorable and significant normalization of histopathological changes in comparison to the FM and FA treated rats (Table 3, Fig. 3B–D).

### 3.5. Changes in gastrointestinal motility

Significant changes in the charcoal transfer rates related to the digestive motility in the SHA control and all three different dosages of HM, compared to the intact control, were absent. However, a significant reduction of the charcoal transfer rate in the gastrointestinal tract in the FM and FA groups, compared to the control groups (P <

**Table 1**
Changes in weight gain and relative organ weights after Hemomine treatment in SHA-induced rats.

| Group | Weight gain (g) | Relative weight (% of body weights) | | |
|---|---|---|---|---|
| | | Spleen | Liver | Left femur |
| Intact-Control | 49.38 ± 5.01 | 0.222 ± 0.032 | 2.960 ± 0.280 | 0.302 ± 0.026 |
| SHA-Control | 21.75 ± 5.44[a] | 0.307 ± 0.014[a] | 2.741 ± 0.133[a] | 0.308 ± 0.024 |
| SHA-FM 0.5 ml/kg | 23.75 ± 2.66[a] | 0.310 ± 0.032[a] | 2.852 ± 0.112 | 0.317 ± 0.023 |
| SHA-FA 1.0 ml/kg | 24.00 ± 6.16[a] | 0.334 ± 0.043[a] | 2.801 ± 0.160 | 0.317 ± 0.022 |
| SHA-HM 2.0 ml/kg | 49.25 ± 6.14[b,c,d] | 0.237 ± 0.023[b,c,d] | 2.957 ± 0.164[b] | 0.303 ± 0.027 |
| SHA-HM 1.0 ml/kg | 40.38 ± 2.97[a,b,c,d] | 0.250 ± 0.027[b,c,d] | 2.852 ± 0.147 | 0.303 ± 0.027 |
| SHA-HM 0.5 ml/kg | 34.25 ± 5.42[a,b,c,d] | 0.265 ± 0.034[a,b,c,d] | 2.874 ± 0.183 | 0.314 ± 0.023 |

Values are expressed Mean ± SD. SHA: Subacute hemorrhagic anemia, FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), HM: Hemomine (60% LJG305).
Different letters indicated $p < 0.05$ as compared with
[a] Intact-Control,
[b] SHA-Control,
[c] SHA-FM,
[d] SHA-FA treated rats by LSD test.

**Table 2**
Hematological values and smear cytology after Hemomine treatment in SHA-induced rats.

| Group | WBC (K/μl) | RBC (M/μl) | Hb (g/dl) | HCT (%) | μRBC (%) | MRBC (%) | PLT (M/μl) | RBC diameters (μm) | PCEs (cells/1000 erythrocytes) |
|---|---|---|---|---|---|---|---|---|---|
| Intact Control | 13.81 ± 2.72 | 6.57 ± 0.32 | 18.98 ± 0.37 | 35.84 ± 1.17 | 1.60 ± 0.29 | 1.68 ± 0.25 | 731.25 ± 90.81 | 7.17 ± 0.61 | 81.13 ± 14.25 |
| SHA-Control | 14.53 ± 3.80 | 4.49 ± 0.34[a] | 13.66 ± 0.53[a] | 25.44 ± 1.08[a] | 3.14 ± 0.27[a] | 2.13 ± 0.22[e] | 939.25 ± 30.56[a] | 4.97 ± 0.50[e] | 424.00 ± 65.55[e] |
| SHA-FM 0.5 ml/kg | 12.38 ± 3.79 | 5.08 ± 0.14[a,b] | 15.48 ± 0.29[a,b] | 30.65 ± 1.49[a,b] | 2.43 ± 0.36[a,b] | 2.40 ± 0.30[e] | 886.38 ± 92.20[a] | 5.57 ± 0.23[e,f] | 273.63 ± 32.11[e,f] |
| SHA-FA 1.0 ml/kg | 15.66 ± 4.63 | 5.50 ± 0.28[a,b] | 16.09 ± 0.70[a,b] | 31.75 ± 1.29[a,b] | 2.39 ± 0.50[a,b] | 2.31 ± 0.42[e] | 839.75 ± 267.27 | 5.93 ± 0.19[e,f] | 247.88 ± 34.09[e,f] |
| SHA-HM 2.0 ml/kg | 13.52 ± 2.78 | 6.20 ± 0.33[a,b,c,d] | 17.61 ± 0.50[a,b,c,d] | 34.51 ± 0.92[a,b,c,d] | 1.76 ± 0.34[b,c,d] | 2.65 ± 0.70[e] | 927.13 ± 100.76[a] | 6.46 ± 0.37[e,f,g,h] | 125.13 ± 15.70[e,f,g,h] |
| SHA-HM 1.0 ml/kg | 12.50 ± 2.74 | 5.96 ± 0.23[a,b,c,d] | 17.24 ± 0.91[a,b,c,d] | 33.14 ± 0.69[a,b,c,d] | 1.84 ± 0.25[b,c,d] | 2.64 ± 0.73[e] | 958.50 ± 123.44[a] | 6.19 ± 0.20[e,f,g,h] | 136.38 ± 18.95[e,f,g,h] |
| SHA-HM 0.5 ml/kg | 13.34 ± 3.24 | 5.73 ± 0.24[a,b,c] | 16.21 ± 0.46[a,b,c] | 32.34 ± 0.97[a,b,c] | 2.03 ± 0.32[a,b,c,d] | 2.46 ± 0.65[e] | 896.50 ± 152.62[a] | 6.02 ± 0.23[e,f,g] | 152.63 ± 17.05[e,f,g,h] |

Values are expressed mean ± SD. SHA: Subacute hemorrhagic anemia, FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), HM: Hemomine (60% LJG305).
WBC: leukocytes, RBC: erythrocytes, Hb: hemoglobin, HCT: hematocrit, μRBC: micro-erythrocyte (small erythrocyte to normal range), MRBC: macro-erythrocyte (larger erythrocyte to normal range), PLT: platelet, PCE: polychromatic erythrocytes
Different letters indicated $p < 0.05$ as compared with
[a] Intact-Control,
[b] SHA-Control,
[c] SHA-FM,
[d] SHA-FA treated rats by LSD test, and with
[e] Intact-Control,
[f] SHA-Control,
[g] SHA-FM,
[h] SHA-FA treated rats by MW test.

0.05) (Table 4), was observed. In addition, there were no changes in the histopathological profiles of the colon and remnant feces *in situ*, mucosa thicknesses, mucous producing cell count, and the surface mucous thicknesses of remnant fecal pellets in the SHA control and all three different dosages of HM treated rats when compared to the intact control. The FM and FA treatment significantly decreased the histopathological profiles of the colon and remnant feces *in situ* when compared to the control groups (Table 4, Fig. 3E).

## 4. Discussion

In this study, we observed that the HM treatment dose-dependently mitigated the abnormal anemic signs in the animal model of anemia, which was induced by subacute bleedings. Especially, the HM treatment had more favorable effects than the equal dosages of FM and FA. In addition, the commercial iron supplements, FM and FA, showed typical constipation signs, such as a reduction in gastrointestinal motilities (Sahebzamani et al., 2013; Souza et al., 2009), whereas the HM administration did not show any notable effects on the gastrointestinal motilities and mucous components on the colon and remnant feces when compared to the control groups. These results are regarded as direct evidence that the HM has potent and favorable anti-anemic effects on SHA, more effective than the commercial iron supplements, FM and FA, and serves as a potential and promising hematopoietic and therapeutic agent for anemia.

Considerable amounts of bleeding causes hypovolemic conditions in mammals, and is accompanied by a marked reduction in body weight (Moghadam et al., 2013; Younes et al., 2007). In this study, the body weight was reduced by blood exsanguinations in the SHA control rats. The hypovolemic related reduction in body weight was dose-dependently mitigated by the treatment of all three different dosages of HM, but not by the FM or FA treatments. It may be suggested that the HM can alleviate the loss in body weight induced by the SHA. Animals with enhanced immune systems exhibited relatively good growth patterns (Duarte et al., 2000; Piao et al., 2003). Previously, it was reported that the positive immunomodulatory effects of the individual or some mixtures contains the following extracts: roots of *Angelica gigas* Nakai (Han et al., 2006), roots of *Paeonia lactiflora* Pall. (Wu et al., 2011), roots of *Rehmannia glutinosa* Liboschitz ex Stueudel (Zhang et al., 2008), rhizomes of *Cnidium officinale* Makino (Kim et al., 2013), or roots of *Glycyrrhiza uralensis* Fischer (Chung et al., 2001). In addition, nodakenin, the major bioactive compound in roots of *Angelica gigas* Nakai, is known to have anti-inflammatory properties (Rim et al., 2012) and is included in HM. Therefore, the inhibitory effects of HM on the SHA related hypovolemic body weight losses might be related to the well-documented immunomodulatory effects of the individual herbal components of LJG305, the main functional materials of the HM.

Studies have shown that iron-deficient and regenerative anemia decreases the levels of RBC, Hb, and HCT, and increases the levels of μRBC, MRBC, and PLT (Joo et al., 2012; Scholbach et al., 2012; Sodikoff, 2001). Furthermore, a reduction in the RBC's diameter and an increase in PCEs were detected in the typical iron-deficient and regenerative anemia (Janus and Moerschel, 2010; Price et al., 2011). In this study, the SHA-related hematological and smear cytological changes were effectively inhibited by the commercial iron supplements, FM and FA, and also by all the three different dosages of HM, dose-dependently, except for PLT numbers. To note, more favorable effects in the hematological inspections were noticed in the HM treatment, compared to the equal dosages of FM and FA.

It was reported that continuous bleeding induced excessive erythropoiesis in the bone marrow while simultaneously starting extramedullary ectopic hematopoiesis in the fetal hematopoietic organs, liver and spleen (Knobel et al., 1993; Melikian et al., 2009). In anemic condition, depletion of the hematopoietic stem cells and reduction of the absolute cell count have been observed with over-differentiation

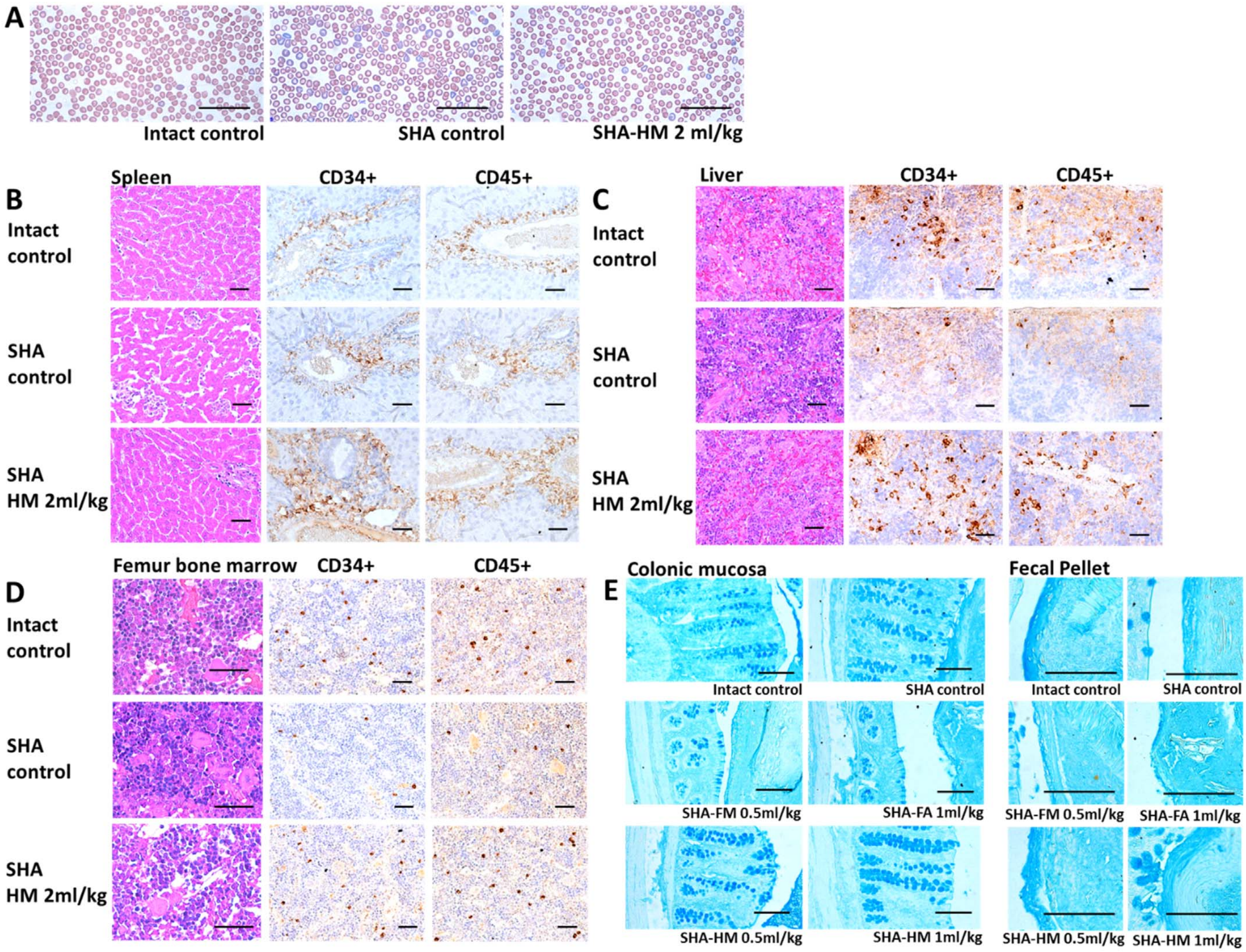


**Fig. 3.** Representative blood smear cytological, histopathological, and immunohistochemistrical images. SHA: Subacute hemorrhagic anemia, FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), HM: Hemomine (60% LJG305). Scale bar: 160 µm. (A) Blood smear cytology. (B) Splenic, (C) Hepatic, (D) Femur bone marrow histopathology. (E) Histochemistrical images of colon and remnant fecal pellet *in situ*.

into erythrocyte precursor cells, or migration into the peripheral blood with premature erythrocytes like MRBC or PCEs (Baranski et al., 2011; Río et al., 2008). In this study, reduction of hematopoietic stem cells was detected in the splenic red pulp, hepatic portal triad regions, and bone marrow of the femur utilizing immunohistochemistry analysis with CD34+ and CD45+ antibodies, which are markers for hematopoietic stem cells (Baranski et al., 2011; Joo et al., 2012; Scholbach et al., 2012). In the SHA control rats, inductions of the splenic red pulp cells, hematopoietic spots on the liver, and hyperplasia of the femur's bone marrow cells with reduction of hematopoietic stem cells, were observed. The ectopic hematopoiesis, hyperplasia of the bone marrow, and SHA-related reduction of the hematopoietic stem cells were inhibited by the treatment of all the test substances. Furthermore, more desirable effects were exhibited in the HM treated rats when compared to the equal dosages of FM and FA. Notably, the HM 1 and 2 ml/kg treatment showed significant inductions of CD34+ and CD45+ cells in the hepatic portal triad and splenic red pulps, compared to the intact control; this observation suggests that the hematopoietic potentials of the HM may be directly correlated with the hyperplasia of the stem cells, rather than the migration inhibitory effects. However, the possibility that the HM may also potentially inhibit the migration of the stem cells could not be completely excluded. Therefore, more fundamental mechanism studies are advised for further study in the future.

The transit process of the entire gastrointestinal tract reflected the overall gastrointestinal motor activity, and measuring the gastrointestinal charcoal transit ratio assists with the diagnosis of constipation (Wintola et al., 2010). In addition, reduction in mucous production in the colon was reported to be correlated to constipation (Yang et al., 2013), and reductions in the colonic mucosa layer thicknesses and mucous producing cells have previously been detected in animals suffering from constipation (McCullough et al., 1998). Therefore, decreased gastrointestinal charcoal transit and mucous components by the FM and FA could be considered as evidence that the commercial iron supplements may induce digestive discomfort, such as constipation (Sahebzamani et al., 2013; Souza et al., 2009). However, the HM treatments had no effects on the charcoal transfers, or in the mucous components in the colonic mucosa and in the remnant feces. This serves as direct evidence that the HM has potent, favorable anti-anemic effects on the SHA with reduced digestive discomfort, which is one of the side effects caused by iron supplements.

There is a limitation of this experiment, which is that there is no further investigation of the underlying mechanisms. Previously, some mixed herbal formulas improved the hematological parameters through the induction of erythropoietin concentration and the activity of aminolevulinic acid dehydrates (Chang et al., 2009; Lee et al., 2014). In addition, iron supplements are known to induce digestive disorders caused by Dysbiosis (Skaar, 2010); however, HM might not affect the composition of the gut microbiota. Further study would be warranted

**Table 3**
Histomorphometrical analysis of spleen, liver, and femur bone marrow after Hemomine treatment in SHA-induced rats.

| Groups | Spleen | | | Liver | | | Femur Bone Marrow | | |
|---|---|---|---|---|---|---|---|---|---|
| | Red pulp cells ($\times10^3$ cells/$mm^2$) | CD34+ cells (cells/$mm^2$) | CD45+ cells (cells/$mm^2$) | Area of hematopoietic spots (%/$mm^2$) | CD34+ cells (cells/$mm^2$) | CD45+ cells (cells/$mm^2$) | Cell numbers ($\times10^3$ cells/$mm^2$) | CD34+ cells (cells/$mm^2$) | CD45+ cells (cells/$mm^2$) |
| Intact Control | 2.84 ± 0.60 | 156.00 ± 20.73 | 135.13 ± 20.62 | 1.62 ± 1.03 | 175.13 ± 13.22 | 281.25 ± 21.20 | 16.96 ± 2.58 | 109.13 ± 23.58 | 127.00 ± 27.13 |
| SHA-Control | 23.32 ± 6.18$^{e}$ | 30.63 ± 10.78$^{e}$ | 28.63 ± 16.04$^{e}$ | 27.10 ± 4.95$^{a}$ | 71.38 ± 10.91$^{e}$ | 95.00 ± 15.71$^{e}$ | 57.39 ± 10.01$^{e}$ | 17.88 ± 2.23$^{e}$ | 37.63 ± 11.15$^{e}$ |
| SHA-FM 0.5 ml/kg | 15.77 ± 2.96$^{e,f}$ | 59.88 ± 13.46$^{e,f}$ | 70.50 ± 14.13$^{e,f}$ | 17.75 ± 3.27$^{a,b}$ | 96.75 ± 11.44$^{e,f}$ | 155.38 ± 11.66$^{e,f}$ | 43.28 ± 4.32$^{e,f}$ | 43.38 ± 10.28$^{e,f}$ | 48.88 ± 6.45$^{e,f}$ |
| SHA-FA 1.0 ml/kg | 13.45 ± 3.20$^{e,f}$ | 86.00 ± 13.04$^{e,f}$ | 99.63 ± 13.84$^{e,f}$ | 16.60 ± 1.98$^{a,b}$ | 109.38 ± 15.55$^{e,f}$ | 188.75 ± 21.24$^{e,f}$ | 40.24 ± 8.34$^{e,f}$ | 46.25 ± 10.98$^{e,f}$ | 61.25 ± 11.36$^{e,f}$ |
| SHA-HM 2.0 ml/kg | 6.62 ± 1.79$^{e,f,g,h}$ | 405.88 ± 123.01$^{e,f,g,h}$ | 318.00 ± 75.52$^{e,f,g,h}$ | 7.66 ± 2.71$^{a,b,c,d}$ | 252.25 ± 50.93$^{e,f,g,h}$ | 317.75 ± 39.37$^{e,f,g,h}$ | 28.05 ± 5.62$^{e,f,g,h}$ | 99.25 ± 14.38$^{f,g,h}$ | 97.63 ± 21.04$^{e,f,g,h}$ |
| SHA-HM 1.0 ml/kg | 7.57 ± 1.28$^{e,f,g,h}$ | 272.50 ± 72.29$^{e,f,g,h}$ | 145.50 ± 31.54$^{f,g,h}$ | 12.18 ± 3.47$^{a,b,c,d}$ | 217.00 ± 26.27$^{e,f,g,h}$ | 228.25 ± 34.45$^{e,f,g,h}$ | 30.76 ± 3.52$^{e,f,g,h}$ | 74.13 ± 10.03$^{e,f,g,h}$ | 83.38 ± 10.99$^{e,f,g,h}$ |
| SHA-HM 0.5 ml/kg | 11.15 ± 1.57$^{e,f,g}$ | 231.25 ± 58.91$^{e,f,g,h}$ | 133.88 ± 17.65$^{f,g,h}$ | 13.08 ± 3.23$^{a,b,c,d}$ | 160.88 ± 16.70$^{f,g,h}$ | 197.50 ± 18.09$^{e,f,g}$ | 34.79 ± 6.20$^{e,f,g}$ | 71.00 ± 15.77$^{e,f,g,h}$ | 75.75 ± 11.34$^{e,f,g,h}$ |

Values are expressed mean ± SD. SHA: Subacute hemorrhagic anemia, FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), HM: Hemomine (60% LJG305).
Different letters indicated $p < 0.05$ as compared with

[a] Intact-Control,
[b] SHA-Control,
[c] SHA-FM,
[d] SHA-FA treated rats by LSD test, and with
[e] Intact-Control,
[f] SHA-Control,
[g] SHA-FM,
[h] SHA-FA treated rats by MW test.

**Table 4**
Charcoal transfers and histomorphometrical analysis of the colon and remnant fecal pellets *in situ* after Hemomine treatment in SHA-induced rats.

| Groups | Charcoal transfer rate[a] (%) | Thickness of colonic mucosa (μm) | Mucosal goblet cells (cells/mm$^2$) | Fecal surface mucous thickness (μm) |
|---|---|---|---|---|
| Intact Control | 74.96 ± 7.76 | 199.93 ± 25.11 | 439.00 ± 120.84 | 31.96 ± 12.02 |
| SHA-Control | 76.38 ± 12.73 | 208.79 ± 32.48 | 458.50 ± 132.02 | 32.56 ± 9.63 |
| SHA-FM 0.5 ml/kg | 54.10 ± 13.28[ab] | 136.14 ± 17.86[ab] | 204.25 ± 109.51[ab] | 13.30 ± 3.01[ef] |
| SHA-FA 1.0 ml/kg | 50.21 ± 9.96[ab] | 94.43 ± 18.52[ab] | 172.25 ± 91.06[ab] | 5.99 ± 1.74[ef] |
| SHA-HM 2.0 ml/kg | 73.37 ± 10.66[cd] | 208.12 ± 28.65[cd] | 446.38 ± 120.04[cd] | 30.76 ± 11.10[gh] |
| SHA-HM 1.0 ml/kg | 78.37 ± 10.08[cd] | 195.85 ± 35.20[cd] | 482.25 ± 133.60[cd] | 31.41 ± 10.09[gh] |
| SHA-HM 0.5 ml/kg | 80.65 ± 6.85[cd] | 203.08 ± 25.21[cd] | 441.50 ± 124.82[cd] | 30.68 ± 6.83[gh] |

Values are expressed Mean ± SD. SHA: Subacute hemorrhagic anemia, FA: Fematin-Asyrup™ (20% ferritin extract glycerin hydrate), FM: Feromax™ (7.14% ferric hydroxide polymaltose complex), HM: Hemomine (60% LJG305).
Different letters indicated $p < 0.05$ as compared with [a]Intact-Control, [b]SHA-Control, [c]SHA-FM, [d]SHA-FA treated rats by LSD test, and with [e]Intact-Control, [f]SHA-Control, [g]SHA-FM, [h]SHA-FA treated rats by MW test.

to investigate the underlining mechanisms of the anti-anemia effects of HM.

## 5. Conclusions

The results suggested that the three different dosages of HM 0.5, 1, and 2 ml/kg significantly inhibited the experimental SHA in rats. In contrast, the results were particularly favorable when compared to the equal dosages of FM and FA. In addition, any dosage of HM did not influence the gastrointestinal motility and the colonic mucous components, whereas the FM and FA intake showed typical constipation signs, such as reduction of the charcoal transfer rates, thickness of the colonic mucosa, and thickness of the fecal surface mucus. Therefore, HM has potent and desirable anti-anemic effects on SHA, is more effective than the commercial iron supplements, FM and FA through proliferating effects on hematopoietic stem cells with less digestive discomfort; hence, HM serves as a promising and new potent hematopoietic and therapeutic agent for anemia.

## Conflict of interests

*The authors have declared that no competing interests exist.*

## Author Contributions

Conceived and designed the experiments: SKK JWK HJL. Performed the experiments: SKK JWK HJL. Analyzed the data: SKK JWK KSK HJL. Contributed reagents/materials/analysis tools: SKK JWK HJL. Wrote the paper: SKK HK JWK KSK HJL. All the authors read and approved the final manuscript.

## Acknowledgment

HJL is supported by a grant (E0164500-01) from the Korea Food Research Institute (http://eng.kfri.re.kr/). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

## References

Akase, T., Akase, T., Onodera, S., Jobo, T., Matsushita, R., Kaneko, M., Tashiro, S., 2003. A comparative study of the usefulness of toki-shakuyaku-san and an oral iron preparation in the treatment of hypochromic anemia in cases of uterine myoma. Yakugaku zasshi: J. Pharm. Soc. Jpn. 123 (9), 817–824.
Baranski, G.M., Offin, M.D., Sifri, Z.C., Elhassan, I.O., Hannoush, E.J., Alzate, W.D., Rameshwar, P., Livingston, D.H., Mohr, A.M., 2011. beta-blockade protection of bone marrow following trauma: the role of G-CSF. J. Surg. Res. 170 (2), 325–331.
Cha, J., Jeong, J., Park, C., Ahn, H., Moon, H., Cho, Y., 2011. Antiobesity activity of fermented Angelicae gigantis by high fat diet-induced obese rats. J. Enzym. Inhib. Med. Chem., 417–430.
Chang, M.S., Kim, D.R., Ko, E.B., Choi, B.J., Park, S.Y., Kang, S.A., Park, S.K., 2009. Treatment with Astragali radix and Angelicae radix enhances erythropoietin gene expression in the cyclophosphamide-induced anemic rat. J. Med. Food 12 (3), 637–642.
Chung, W.T., Lee, S.H., Kim, J.D., Sung, N.S., Hwang, B., Lee, S.Y., Yu, C.Y., Lee, H.Y., 2001. Effect of the extracts from Glycyrrhiza uralensis Fisch on the growth characteristics of human cell lines: anti-tumor and immune activation activities. Cytotechnology 37 (1), 55–64.
Cook, J.D., 2005. Diagnosis and management of iron-deficiency anaemia. Best. Pract. Res. Clin. Haematol. 18 (2), 319–332.
Coyne, D.W., Kapoian, T., Suki, W., Singh, A.K., Moran, J.E., Dahl, N.V., Rizkala, A.R., Group, D.S., 2007. Ferric gluconate is highly efficacious in anemic hemodialysis patients with high serum ferritin and low transferrin saturation: results of the Dialysis Patients' Response to IV Iron with Elevated Ferritin (DRIVE) Study. J. Am. Soc. Nephrol. 18 (3), 975–984.
Demers, G., Griffin, G., De Vroey, G., Haywood, J.R., Zurlo, J., Bédard, M., 2006. Harmonization of animal care and use guidance. Science 312 (5774), 700–701.
Diehl, K.H., Hull, R., Morton, D., Pfister, R., Rabemampianina, Y., Smith, D., Vidal, J.M., van de Vorstenbosch, C., European Federation of Pharmaceutical Industries, A., European Centre for the Validation of Alternative, M., 2001. A good practice guide to the administration of substances and removal of blood, including routes and volumes. J. Appl. Toxicol.: JAT 21 (1), 15–23.
Duarte, C.G., dos Santos, G.L., Azzolini, A.E.C.S., de Assis Pandochi, A.I., 2000. The effect of the antithyroid drug propylthiouracil on the alternative pathway of complement in rats. Int. J. Immunopharmacol. 22 (1), 25–33.
Feltrin, C., De Morais, M.B., de Cássia Freitas, K., De Morais, T.B., Neto, U.F., Amancio, O.M.S., 2009. Effect of soluble fiber pectin on growth and intestinal iron absorption in rats during recovery from iron deficiency anemia. Biol. Trace Elem. Res. 129 (1–3), 221–228.
Geisser, P., 2007. Safety and efficacy of iron (III)-hydroxide polymaltose complex. Arzneimittelforschung 57 (06), 439–452.
Han, S.-B., Lee, C.W., Kang, M.R., Yoon, Y.D., Kang, J.S., Lee, K.H., Yoon, W.K., Lee, K., Park, S.-K., Kim, H.M., 2006. Pectic polysaccharide isolated from Angelica gigas Nakai inhibits melanoma cell metastasis and growth by directly preventing cell adhesion and activating host immune functions. Cancer Lett. 243 (2), 264–273.
Ihedioha, J.I., Daniel-Igwe, G., 2014. Development of an experimental albino rat (Rattus norvegicus) model for studies on sub-acute blood loss anaemia. Anim. Res. Int. 11 (1), 1889.
Janus, J., Moerschel, S.K., 2010. Evaluation of anemia in children. Am. Fam. Physician 81 (12), 1462–1471.
Jo, S., Park, H., Jung, U., Oh, H., Kim, S., Yee, S., 2005. Protective effect of a herbal preparation (HemoHIM) on the self-renewal tissues and immune system against γ-irradiation. J. Korean Soc. Food Sci. Nutr..
Joo, S.-Y., Chung, Y.S., Choi, B., Kim, M., Kim, J.-H., Jun, T.-G., Chang, J., Sprent, J., Surh, C.D., Joh, J.-w., 2012. Systemic human T cell developmental processes in humanized mice cotransplanted with human fetal thymus/liver tissue and hematopoietic stem cells. Transplantation 94 (11), 1095–1102.
Kassebaum, N.J., Jasrasaria, R., Naghavi, M., Wulf, S.K., Johns, N., Lozano, R., Regan, M., Weatherall, D., Chou, D.P., Eisele, T.P., 2014. A systematic analysis of global anemia burden from 1990 to 2010. Blood 123 (5), 615–624.
Kim, D.H., Kim, Y.C., Jung, J.W., Lee, S., Yoon, B.H., Cheong, J.H., Kim, Y.S., Kang, S.S., Ko, K.H., Ryu, J.H., 2007. Nodakenin, a coumarin compound, ameliorates scopolamine-induced memory disruption in mice. Life Sci. 80 (21), 1944–1950.
Kim, J.-J., Cho, H.W., Park, H.-R., Jung, U., Jo, S.-K., Yee, S.-T., 2013. Preventative effect of an herbal preparation (HemoHIM) on development of airway inflammation in mice via modulation of Th1/2 cells differentiation. PloS One 8 (7), e68552.
Kim, Y.-J., Park, S.-J., Kim, T.-J., 2011. Anti-allergic effects of nodakenin in IgE/Ag-induced type I hypersensitivity. J. Life Sci. 21 (12), 1721–1725.
Knobel, B., Melamud, E., Virag, I., Meytes, D., 1993. Ectopic medullary hematopoiesis as a cause of ascites in agnogenic myeloid metaplasia. Acta Haematol. 89 (2), 104–107.
Koury, M.J., Ponka, P., 2004. New insights into erythropoiesis: the roles of folate, vitamin B12, and iron. Annu. Rev. Nutr. 24, 105–131.
Lee, H., Blaufox, M., 1985. Blood volume in the rat. J. Nucl. Med. 26 (1), 72–76.
Lee, H.W., Kim, H., Ryuk, J.A., Kil, K.-J., Ko, B.S., 2014. Hemopoietic effect of extracts from constituent herbal medicines of Samul-tang on phenylhydrazine-induced hemolytic anemia in rats. Int. J. Clin. Exp. Pathol. 7 (9), 6179.
Lee, S., Shin, D.-S., Kim, J.S., Oh, K.-B., Kang, S.S., 2003. Antibacterial coumarins

fromAngelica gigas roots. Arch. pharmacal Res. 26 (6), 449–452.
Levene, A., 1981. Pathological factors influencing excision of tumours in the head and neck. Part I. Clin. Otolaryngol. Allied Sci. 6 (2), 145–151.
McCullough, J., Ratcliffe, B., Mandir, N., Carr, K., Goodlad, R., 1998. Dietary fibre and intestinal microflora: effects on intestinal morphometry and crypt branching. Gut 42 (6), 799–806.
Melikian, A., Egorova, E., Karagiulian, S., Silaev, M., Kaplanskaia, I., Gitis, M., Kolodeĭ, S., Kovaleva, L., 2009. [Hereditary spherocytic hemolytic anemia in an adult with the formation of ectopic foci of extramedullary hemopoiesis in the chest]. Ter. arkhiv 82 (7), 72–75.
Moghadam, A.M., Natanzi, M.M., Djalali, M., Saedisomeolia, A., Javanbakht, M.H., Saboor-Yaraghi, A.A., Zareei, M., 2013. Relationship between blood donors' iron status and their age, body mass index and donation frequency. Sao Paulo Med. J. 131 (6), 377–383.
Nakamoto, H., Mimura, T., Honda, N., 2008. Orally administrated Juzen-taiho-to/TJ-48 ameliorates erythropoietin (rHuEPO)-resistant anemia in patients on hemodialysis. Hemodial. Int. 12 (s2), S9–S14.
Piao, J.-H., Zhang, L., Zhang, H., Gao, T.-H., Li, X.-H., 2003. [Experimental studies on antianimia effect of shengxuesu]. Zhongguo Zhong yao za zhi= Zhongguo zhongyao zazhi= China J. Chin. Mater. Med. 28 (6), 544–547.
Price, E.A., Mehra, R., Holmes, T.H., Schrier, S.L., 2011. Anemia in older persons: etiology and evaluation. Blood Cells, Mol. Dis. 46 (2), 159–165.
Pulina, M.O., Sokolov, A.V., Zakharova, E.T., Kostevich, V.A., Vasilyev, V.B., 2010. Effect of lactoferrin on consequences of acute experimental hemorrhagic anemia in rats. Bull. Exp. Biol. Med. 149 (2), 219–222.
Rim, H.-K., Cho, W., Sung, S.H., Lee, K.-T., 2012. Nodakenin suppresses lipopolysaccharide-induced inflammatory responses in macrophage cells by inhibiting tumor necrosis factor receptor-associated factor 6 and nuclear factor-κB pathways and protects mice from lethal endotoxin shock. J. Pharmacol. Exp. Ther. 342 (3), 654–664.
Río, P., Meza, N.W., González-Murillo, Á., Navarro, S., Álvarez, L., Surrallés, J., Castella, M., Guenechea, G., Segovia, J.C., Hanenberg, H., 2008. In vivo proliferation advantage of genetically corrected hematopoietic stem cells in a mouse model of Fanconi anemia FA-D1. Blood 112 (13), 4853–4861.
Sagar, L., Sehgal, R., Ojha, S., 2005. Evaluation of antimotility effect of Lantana camara L. var. acuelata constituents on neostigmine induced gastrointestinal transit in mice. BMC Complement. Altern. Med. 5 (1), 1.
Sahebzamani, F.M., Berarducci, A., Murr, M.M., 2013. Malabsorption anemia and iron supplement induced constipation in post-Roux-en-Y gastric bypass (RYGB) patients. J. Am. Assoc. Nurse Pract. 25 (12), 634–640.
Scholbach, J., Schulz, A., Westphal, F., Egger, D., Wege, A.K., Patties, I., Köberle, M., Sack, U., Lange, F., 2012. Comparison of hematopoietic stem cells derived from fresh and cryopreserved whole cord blood in the generation of humanized mice. PloS One 7 (10), e46772.
Sharma, A., Fish, B.L., Moulder, J.E., Medhora, M., Baker, J.E., Mader, M., Cohen, E.P., 2014. Safety and blood sample volume and quality of a refined retro-orbital bleeding technique in rats using a lateral approach. Lab Anim. 43 (2), 63.
Shi, S.-R., Chaiwun, B., Young, L., Cote, R., Taylor, C., 1993. Antigen retrieval technique utilizing citrate buffer or urea solution for immunohistochemical demonstration of androgen receptor in formalin-fixed paraffin sections. J. Histochem. Cytochem. 41 (11), 1599–1604.
Skaar, E.P., 2010. The battle for iron between bacterial pathogens and their vertebrate hosts. PLoS Pathog. 6 (8).
Sodikoff, C.H., 2001. Laboratory Profiles of Small Animal Diseases: A Guide to Laboratory Diagnosis. Mosby Inc.
Son, C.-G., Han, S.-H., Cho, J.-H., Shin, J.-W., Cho, C.-H., Lee, Y.-W., Cho, C.-K., 2003. Induction of hemopoiesis by saenghyuldan, a mixture of ginseng radix, Paeoniae radix alba, and Hominis placenta extracts. Acta Pharmacol. Sin. 24 (2), 120–126.
Souza, A.Id, Batista Filho, M., Bresani, C.C., Ferreira, L.O.C., Figueiroa, J.N., 2009. Adherence and side effects of three ferrous sulfate treatment regimens on anemic pregnant women in clinical trials. Cad. De. Saúde Pública 25 (6), 1225–1233.
Szarfarc, S.C., De Cassana, L., Fujimori, E., Guerra-Shinohara, E.M., de Oliveira, I., 2001. Relative effectiveness of iron bis-glycinate chelate (Ferrochel) and ferrous sulfate in the control of iron deficiency in pregnant women. Arch. Latinoam. De. Nutr. 51 (1Suppl 1), 42–47.
Tang, X., Falls, D.L., Li, X., Lane, T., Luskin, M.B., 2007. Antigen-retrieval procedure for bromodeoxyuridine immunolabeling with concurrent labeling of nuclear DNA and antigens damaged by HCl pretreatment. J. Neurosci. 27 (22), 5837–5844.
Wintola, O.A., Sunmonu, T.O., Afolayan, A.J., 2010. The effect of Aloe ferox mill. in the treatment of loperamide-induced constipation in Wistar rats. BMC Gastroenterol. 10 (1), 95.
Wu, D., Zhou, J., Wang, X., Cui, B., An, R., Shi, H., Yuan, J., Hu, Z., 2011. Traditional Chinese formula, lubricating gut pill, stimulates cAMP-dependent Cl− secretion across rat distal colonic mucosa. J. Ethnopharmacol. 134 (2), 406–413.
Xu, J., 1992. [Research on liu wei Rehmannia oral liquid against side-effect of drugs of anti-tumor chemotherapy]. Zhongguo Zhong xi yi jie he za zhi Zhongguo Zhongxiyi jiehe zazhi= Chinese journal of integrated traditional and Western medicine/ Zhongguo Zhong xi yi jie he xue hui, Zhongguo Zhong yi yan jiu yuan zhu ban 12(12), 734-737, 709-710.
Yang, B., Xiao, B., Sun, T., 2013. Antitumor and immunomodulatory activity of Astragalus membranaceus polysaccharides in H22 tumor-bearing mice. Int. J. Biol. Macromol. 62, 287–290.
You, K., Wu, B., Huang, Z.Q., Zhang, C., Pan, L., Wu, G., Zhao, Z., Huang, M., 1996. [Effect of hemin in treating hemorrhagic anemia and toxicity]. Zhongguo yao li xue bao= Acta Pharmacol. Sin. 17 (3), 284–286.
Younes, R.N., Deutsch, F., Itinoshe, M., Fontes, B., Poggetti, R., Birolini, D., 2007a. Responsiveness to different volume therapies following hemodilution and hemorrhagic shock: a comparative experimental study in rats. Acta Cir. Bras. 22 (5), 355–360.
Zee-Cheng, R., 1992. A potent Chinese biological response modifier in cancer immunotherapy, potentiation and detoxification of anticancer drugs. Methods Find. Exp. Clin. Pharm. 14, 725–736.
Zhang, R.-X., Li, M.-X., Jia, Z.-P., 2008. Rehmannia glutinosa: review of botany, chemistry and pharmacology. J. Ethnopharmacol. 117 (2), 199–214.